AI Assisted Coding
Lab Assignment 13.4

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B14

Task 1:RefactoringDataTransformationLogic

TaskDescription:

• Reviewthelegacycodethatcomputestransformedvaluesusing an

explicit loop.

• UseanAItooltosuggestamorePythonicrefactoringapproach.

• Refactorthecodeusinglistcomprehensionsorhelperfunctions

while preserving the output.

PROMPT:

RefactorthefollowingPythoncodetomakeitshorterandmorePythonicusinglist comprehension while keeping the same output.

values=[2,4,6,8,10] doubled =

[]

forvinvalues:

doubled.append(v*2)

print(doubled)

Task 2:ImprovingTextProcessingCodeReadability

• Analyze the legacy code that constructs a sentence using

repeated string concatenation.

• Ask AIto suggest a more efficient and readable approach.

Refactor the codeaccordingly while keepingthe final output unchanged.

PROMPT:

Refactor the following Python code to improve readability and efficiency by avoiding repeated string concatenation. Keep the same output.

```
AI ASS LAB_13.4.py U
AI ASS LAB_13.4.py > ...
words = ["Refactoring", "with", "AI", "improves", "quality"]
message = " ".join(words)
print(message)
```

OUTPUT:

```
PS C:\Users\chunc_yhjtd63\OneDrive\Desktop\AI ASS COD> & C:\Users\chunc_yhjtd63\AppData\Local\Programs\Python\Python313\python.exe
"c:/Users/chunc_yhjtd63/OneDrive/Desktop/AI ASS COD/AI ASS LAB_13.4.py"
Refactoring with AI improves quality
PS C:\Users\chunc_yhjtd63\OneDrive\Desktop\AI ASS COD>
```

Task3:Safer Access to Configuration Data

• Review thelegacycodethatmanuallychecksfordictionarykeys.

• UseAIsuggestionstorefactorthecodeusingsaferdictionary

access methods.

• Ensurethebehaviorremainsthesameformissingkeys.

PROMPT:

RefactorthefollowingPythoncodetouseasaferdictionaryaccessmethodinsteadof manually checking keys. Keep the same output if the key is missing.

config={"host":"localhost","port":8080} if

"timeout" in config:

print(config["timeout"])

else:

print("Defaulttimeoutused")

```
AI ASS LAB_13.4.py U
AI ASS LAB_13.4.py > ...
config = {"host": "localhost", "port": 8080}
print(config.get("timeout", "Default timeout used"))
```

OUTPUT:

```
PS C:\Users\chunc_yhjtd63\OneDrive\Desktop\AI ASS COD> & C:\Users\chunc_yhjtd63\AppData\Local\Programs\Python\Python313\python.ex
"c:/Users/chunc_yhjtd63/OneDrive/Desktop/AI ASS COD/AI ASS LAB_13.4.py"
Default timeout used
```

Task4:Refactoring Conditional Logic for Scalability

Examinethemultipleif–elifconditionsusedtodetermine operations.

AskAItosuggestacleaner,scalablealternative.

Refactorthelogicusingmappingtechniqueswhilepreserving

functionality.

PROMPT:

RefactorthefollowingPythoncodetoreplacemultipleif–elifconditionswithacleanerandscalable mapping approach while keeping the same output.

```
AI ASS LAB_13.4.py U
AI ASS LAB_13.4.py > action
action = "divide"
x, y = 10, 2

operations = {
    "add": lambda a, b: a + b,
    "subtract": lambda a, b: a - b,
    "multiply": lambda a, b: a * b,
    "divide": lambda a, b: a / b,
}

result = operations.get(action, lambda a, b: None)(x, y)
print(result)
```

OUTPUT:

```
PS C:\Users\chunc_yhjtd63\OneDrive\Desktop\AI ASS COD> & C:\Users\chunc_yhjtd63\AppData\Local\Programs\Python\Python313\python.exe
"c:/Users/chunc_yhjtd63/OneDrive/Desktop/AI ASS COD/AI ASS LAB_13.4.py"
5.0
PS C:\Users\chunc_yhjtd63\OneDrive\Desktop\AI ASS COD>
```

Task5:SimplifyingSearchLogicinCollections

Identifytheexplicitloopusedforsearchinganitem.

UseAIassistancetorefactorthelogicintoamoreconciseand readable

form.

Maintainthesameoutputbehavior.

PROMPT:

RefactorthefollowingPythoncodetosimplifythesearchlogicinacollectionusingamoreconcisePython approach while keeping the same output.

```
AI ASS LAB_13.4.py U
AI ASS LAB_13.4.py > inventory
inventory = ["pen", "notebook", "eraser", "marker"]
found = "eraser" in inventory
print("Item Available" if found else "Item Not Available")
```

OUTPUT:

```
PS C:\Users\chunc_yhjtd63\OneDrive\Desktop\AI ASS COD> & C:\Users\chunc_yhjtd63\AppData\Local\Programs\Python\Python313\python.exe
"c:/Users/chunc_yhjtd63/OneDrive/Desktop/AI ASS COD/AI ASS LAB_13.4.py"
Item Available
PS C:\Users\chunc_yhjtd63\OneDrive\Desktop\AI ASS COD>
```